# PROJECT PROPOSAL

## Explainable Multimodal Deep Learning for Medical Diagnosis Using Chest X-ray Images and Clinical Text

**Submitted by:**

| Name | Reg Number |
|---|---|
| Toqeer Ahmed | 2023728 |
| Muneeb Bin Nasir | 2022463 |


**Course: Deep Learning**

## 1. PROBLEM STATEMENT

Most existing deep learning-based diagnostic systems are unimodal--they process either chest X-ray images or clinical text reports, but not both. This limits their diagnostic accuracy since each modality captures complementary information. Moreover, these models function as black boxes, offering predictions without any explanation, which makes them unsuitable for clinical deployment where transparency is essential. This project addresses both limitations by building a multimodal system that jointly processes images and text, and provides interpretable, human-understandable explanations for each prediction.

## 2. MOTIVATION

Chest diseases such as pneumonia, pleural effusion, and atelectasis are among the leading causes of hospital admissions worldwide. Radiologists routinely combine visual findings from X-rays with clinical notes to reach a diagnosis--yet AI systems rarely replicate this practice. Multimodal learning offers a path to bridging this gap. Furthermore, clinical adoption of AI tools is hindered not only by accuracy concerns but by the lack of interpretability. Clinicians need to understand why a model made a decision before they can act on it. Explainable AI (XAI) methods such as Grad-CAM and attention visualization address this need directly, making the model's reasoning visible and verifiable.

## 3. BACKGROUND

Deep learning has demonstrated remarkable performance in medical image analysis. DenseNet-121 (Huang et al., 2017), in particular, has become a standard backbone for chest X-ray classification after CheXNet (Rajpurkar et al., 2017) demonstrated radiologist-level pneumonia detection. In parallel, Transformer-based language models--especially BERT (Devlin et al., 2019) and its clinical adaptation ClinicalBERT (Alsentzer et al., 2019)--have proven effective for processing radiology reports and clinical notes. Multimodal fusion strategies range from simple concatenation (early/late fusion) to cross-modal attention mechanisms that allow each modality to query the other, yielding richer joint representations (Chen et al., 2020). For explainability, Grad-CAM (Selvaraju et al., 2017) generates class-discriminative heatmaps over image regions, while BERT attention weights highlight influential tokens in the text--together forming a complete explanation layer for multimodal predictions.

## 4. RELATED WORK

Rajpurkar et al. (2017) established the viability of CNN-based chest X-ray classification with CheXNet, while Wang et al. (2018) released the ChestX-ray14 benchmark enabling large-scale multi-label disease classification--both operating solely in the image domain. Johnson et al. (2019) published MIMIC-CXR, pairing over 227,000 chest X-ray studies with free-text reports, enabling multimodal research at scale. Huang et al. (2021) subsequently demonstrated that fusing DenseNet image features with ClinicalBERT text embeddings via co-attention improved classification over unimodal baselines. On the XAI side, Selvaraju et al. (2017) introduced Grad-CAM as a reliable localization tool, later validated in pneumonia detection by Pawar et al. (2020). Despite these advances, a unified system combining multimodal fusion with synchronized image-text explanations remains underexplored—the gap this project targets.

## 5. PROPOSED METHODOLOGY

The system consists of three main components. The visual encoder uses DenseNet-121 pre-trained on ImageNet, fine-tuned on chest X-ray images resized to 224×224 pixels. Spatial feature maps from the final convolutional block are extracted and passed to the fusion module. The text encoder uses ClinicalBERT to process radiology reports, with token-level embeddings retained for both fusion and attention visualization. Fusion is performed via a cross-modal attention mechanism: image patch embeddings and BERT token embeddings are projected into a shared 512-dimensional space, and a multi-head attention block allows each modality to attend to the other. The resulting attended features are concatenated and passed through a fully connected classification head with sigmoid output for 14 thoracic disease labels, trained with weighted binary cross-entropy loss to handle class imbalance.

For explainability, Grad-CAM is applied to the final DenseNet convolutional layer to generate heatmaps overlaid on the original X-ray, highlighting pathology-relevant regions. Simultaneously, cross-modal attention weights are extracted and mapped back to report tokens to identify the most diagnostically significant phrases. Both explanations are generated together for every prediction. The system will be trained and evaluated on MIMIC-CXR (primary) and Open-I (secondary validation). Implementation uses Python, PyTorch, HuggingFace Transformers, OpenCV, and Captum. Performance will be measured using AUC-ROC (per class and macro-averaged), F1-score, Precision, and Recall, with explanation quality assessed via IoU against radiologist bounding box annotations.

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